CHOCKALINGAM BALAN

Full-Stack Data & AI Engineer — Databricks, PySpark, Backend & LLM Systems

Chennai, India · +91-9524031307 · chockalingam1805@gmail.com · Portfolio · LinkedIn · GitHub

PROFESSIONAL SUMMARY

Full-stack data and AI engineer with 6+ years across data platforms, backend services, and applied AI. Strongest in Databricks and PySpark lakehouse work (Delta Lake, Unity Catalog, medallion, Parquet) backed by solid SQL, dimensional modeling, and data validation, and equally hands-on shipping Django and FastAPI services, React front ends, and LLM-powered workflows with OpenAI, Claude, and AWS Bedrock. Known for replacing slow manual processes with reusable, automated accelerators, and for working directly with client and business teams from requirements and sprint demos through production rollout. Have led engineering teams across backend, data, and ML workstreams, and built multi-tenant platforms serving 10–12 international markets.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, TypeScript, Golang (familiar)
Backend	FastAPI, Django, Flask, REST API Design, Microservices, Async Processing (Celery, Redis), Event-Driven Architecture
Data Platform	Databricks & PySpark (6+ years), Delta Lake, Unity Catalog, Medallion Architecture (Bronze/Silver/Gold), Dimensional Modeling (Fact/Dimension), Parquet, Data Quality & Validation, Databricks Genie, Databricks SDK, Databricks Asset Bundles (DAB), Kafka, Hive, ETL Pipelines
BI & Warehouses	Power BI (semantic models & dashboards), Microsoft Fabric (familiar), Snowflake (familiar), Snowflake Cortex (familiar), BigQuery (exposure)
AI / ML	OpenAI API, Claude (Sonnet/Opus), AWS Bedrock, LLM Integration, Streamlit, AI-Powered Code Generation, Time Series Forecasting, Regression, Stacked Ensembles, AutoML, PyTorch, TensorFlow, Keras, XGBoost, LightGBM, H2O.ai
Frontend	React.js, Angular, Spring Boot, Next.js (familiar), TypeScript
Databases	SQL Server, PostgreSQL, MySQL, Databricks Hive, MongoDB (familiar)
Cloud / Infra	Azure, GCP, AWS, Docker, AKS, CI/CD Pipelines, Monitoring & Observability
Orchestration	Apache Airflow, Celery, Agile/Scrum, AI-Assisted Development Tools

PROFESSIONAL EXPERIENCE

Technical Specialist (Data & AI Engineering Lead)

Dec 2025 – Present

iLink Digital Inc · Chennai, India

Led and mentored a team of 10 engineers across backend, data engineering, and AI workstreams — driving architecture decisions, sprint planning, and code reviews. Customer-facing on BMS and AI Claims: handled direct client communication, stakeholder sprint planning, and rapid iteration.

BMS — Enterprise BI Migration Platform (Customer-Facing) | *Python, FastAPI, Azure Databricks, PySpark, SQL Server, Power BI, OpenAI, Claude, Bedrock*

- **Databricks notebook development:** Built 10–15 Databricks notebooks linking 20+ data sources (e.g. Vendor Source List, Material Supplier Master with Last PO & Invoice Price, Manual Projection Report), with validation and data quality checks; delivered to UAT.
- **Power BI semantic models:** Developed 5–10 Power BI semantic models (e.g. HR_CASE, PO List, PDL and ATL Users) on top of the migrated Databricks data for business reporting; delivered to UAT.
- **Migration accelerators:** Cut Tableau-to-Databricks migration time from 5–14 days to 10–15 minutes per flow (99%+ faster, 80–85% automation accuracy) across 20+ data sources by architecting the Flow Migrator (Tableau Prep flows to Databricks notebooks or Power BI M Query: flow parsing, execution-order replication, dependency analysis, topological sorting, intelligent chunking, node-based routing) and the TDS Migrator

(Tableau data sources to notebooks or M Query: XML parsing, metadata extraction, dependency resolution, automated code generation).

- **LLM integration:** Integrated OpenAI, Claude (Sonnet/Opus), and AWS Bedrock with controlled, optimized LLM orchestration to interpret complex Tableau transformation logic, metadata, dependencies, and business rules for automated code generation at scale; engineered an OpenAI-based fallback execution engine for unsupported or complex transformations that could not be parsed deterministically, significantly improving coverage.
- **Transformation engine:** Engineered transformation parsing and compilation layer converting Tableau operations (joins, unions, aggregations, calculated fields, filters, nested expressions) into executable Spark and Pandas transformations.
- **Throughput & orchestration:** Scaled team delivery throughput from 5–7 to 40–50 migrations per week (5x+) by architecting a DAG-based ETL execution engine with dependency resolution and topological sorting, plus reusable AI-powered pipelines; built the FastAPI orchestration service for the full workflow lifecycle (ingestion, DAG generation, execution triggering, monitoring) with async concurrent execution and API response times under 5 seconds.
- **Validation framework:** Cut developer data validation from 10–12 hours to 10–30 minutes (95%+ faster) with the BMS Unit Test Accelerator, an AI-powered validation framework that compares migrated Tableau logic with generated Databricks/M Query outputs across metadata, schema, datatype, row count, and data accuracy, using deterministic tests to flag discrepancies and eliminate manual validation effort.
- **Platform & tooling:** Integrated Azure Databricks as the distributed execution engine and Azure Blob Storage for input/output artifacts; designed the SQL Server schema for workflow metadata, execution status, and tracking; used Databricks Genie for pipeline development and notebook exploration and the Databricks SDK for programmatic workspace and job management.
- **Reusable architecture:** Built BMS migration pipelines as modular, reusable components following Agent Development Kit (ADK) patterns — enabling team-wide reuse rather than isolated implementations; planned product features and roadmap updates directly with client business and product teams in Agile sprint cadence.
- **Lineage, optimization & assessment:** Built the Lineage accelerator (upstream and downstream dependencies and execution sequencing across flows and data sources), the M Query Optimizer/Normalizer (SQL pruning, column reduction, query normalization for better runtime and lower memory), and the Connection Inventory accelerator (source connections and dependencies for migration assessment).
- **Stakeholder management:** Coordinated directly with client stakeholders in the absence of a dedicated lead: ran discussions, clarified requirements, gave project updates, and addressed concerns to keep delivery on track; also ran short R&D on Power Automate, the EDP Framework, and SharePoint connectors.

AI Claims Adjudication Platform (Customer-Facing POC) | *Python, FastAPI, Docker, LLM*

- Cut claim adjudication response time from 25s to 10s (60% faster) by refactoring the backend processing pipeline and implementing async concurrent execution — discussed directly with the customer as a POC with iterative Agile changes.
- Designed and owned FastAPI backend for insurance claim ingestion and AI-based adjudication orchestration: structured extraction, evaluation, and classification of claim data; configurable request/response schemas; AI inference workflow; Docker containerization for deployment portability.

Nucor — Enterprise React Platform Refactoring | *React, Suspense*

- Redesigned routing architecture with protected routes, lazy loading, and React Suspense — reducing initial load time and securing access control; modularized session storage and application state handling improving maintainability.

Machine Learning Engineer

Jan 2025 – Nov 2025

Tiger Analytics · Chennai, India

Led a team of 3 engineers across backend, ETL, and ML model deployment. Delivered sprint-wise demos of reports and backend features directly to client business teams, product teams, and stakeholders.

PepIris — Retail Computer Vision Mobile Platform (Customer-Facing) | *Django, Celery, Azure Databricks, PySpark, Delta Lake, SQL Server*

- Designed Django multi-tenant backend deployed across 10–12 international markets from a single shared codebase with full tenant schema isolation — zero cross-market impact on deployments or schema changes.

- Built scalable REST APIs supporting mobile application workflows, analytics retrieval, and reporting; implemented Celery-based async task execution framework for scalable ML inference and background processing.
- Reduced QA validation cycle time by 90%+ by building a direct model validation workflow and automated ML deployment pipeline with dynamic credential and version management for runtime model switching without redeployment.
- Migrated legacy Pandas ETL pipelines to distributed PySpark on Azure Databricks with Delta Lake, Unity Catalog, and a bronze/silver/gold medallion architecture — used Databricks Genie for pipeline development and data exploration, and Databricks Asset Bundles (DAB) for deployment packaging and environment promotion — with configuration-driven reuse across all international markets; led production go-live rollouts across multiple markets.
- Consolidated 30+ disparate report and data-load notebooks into a single reusable, parameter-driven repository, improving pipeline governance and maintainability across markets; led three production deployments and served as technical owner for production rollouts in 3–4 markets.
- Designed and maintained SQL Server tables for new features (Business Rules, Master Data Management); used Celery with Redis as broker for asynchronous model deployment and data-validation tasks; contributed React Native changes to the mobile app.

Senior Analyst

Oct 2023 – Dec 2024

Tiger Analytics · Chennai, India

Peplris — Retail Computer Vision Mobile Platform | *Django REST, SQL Server, Pandas, SQLAlchemy*

- Built Django REST APIs with versioning for backward-compatible feature rollouts; designed multi-market SQL Server schema; implemented database connection pooling and Pandas/SQLAlchemy ETL pipelines for scalable data ingestion.
- Optimized database queries and connection pooling improving backend performance and API response time across 10–12 international markets.
- Automated data-load and report pipelines, replacing manual processes and improving report efficiency by 45%; tuned Django ORM queries, standardized API response and error handling, and configured ODBC/pyodbc drivers for higher data throughput.
- Containerized Django services with Docker and Gunicorn for environment parity across markets; designed endpoint versioning on a common repository shared by 8–15 markets.

Fullstack & AI Application Developer

Nov 2020 – Oct 2023

KAAR Technologies · Chennai, India

Led a team of 5 developers across fullstack ML platforms. Personally reviewed model outputs and delivered analysis directly to business teams as Power BI reports with follow-up discussion.

Model Builder — ML Training Platform (Customer-Facing) | *React, Flask, Airflow, H2O.ai, XGBoost, LightGBM, Power BI*

- Built end-to-end ML platform handling large datasets with React frontend and Flask backend for model configuration, training, monitoring, and lifecycle management; integrated H2O.ai AutoML and engineered custom AutoML pipelines (XGBoost, LightGBM, Gradient Boosting) reducing dependency on proprietary tools.
- **Forecasting & AutoML:** Built regression-based time series forecasting models, plus stacked ensemble and custom AutoML models (PyTorch, Keras, TensorFlow, XGBoost, LightGBM) that replicate H2O.ai behavior, reaching 75–80% accuracy (vs ~90% for H2O.ai) on the Model and Scenario Builder use cases without a proprietary-tool dependency.
- Designed Apache Airflow DAGs for automated model training, prediction, retraining, and evaluation workflows; personally reviewed baseline model runs, selected best-performing runs, and delivered analysis directly to business teams via Power BI reports with follow-up discussion.
- **Data modeling:** Modeled sales data into fact and dimension tables for forecasting on Databricks, using Parquet, Delta Lake and bronze/silver/gold medallion layers.
- **Dashboards:** Built model run dashboards in Model Builder and Scenario Builder for month-wise comparisons and cross-comparison of Model Builder and Scenario Builder runs.

Scenario Builder — Forecast Simulation Platform (Customer-Facing) | *React, Flask, Airflow, PySpark, Databricks Hive, GCP, Firestore*

- Delivered customer-facing what-if forecasting web app for scenario-based ML simulation (COVID cases +100%, footfall +20%, single or combined events) — React + Flask + Airflow + PySpark on Databricks Hive + GCP Bucket + Firestore — used directly by business teams to determine KPIs alongside Model Builder baseline forecasts.
- Implemented async execution and Airflow DAGs for automated and manual workflows; personally reviewed scenario forecasts and delivered best-run analysis to business teams.

AI Web Applications — Demand Planning, Forecasting & Master Data Management | *React, Angular 7, Flask, Spring Boot, PostgreSQL 11, Azure AD, Power BI*

- Developed AI web applications with Azure AD authentication and role-based access: configuration create/edit/share/run, Power BI results and file downloads, admin screens and user-group-specific views; built Angular 7 UI with Spring Boot APIs on PostgreSQL 11 for Master Data Management, Demand Planning and Forecasting & Inventory Allocation modules, integrating Python ML models.
- Ran code reviews, maintained three web apps, and delivered two on time using Azure DevOps.

OPEN SOURCE & PROJECTS

Schema Shield (open source, on PyPI) github.com/CHOCKA-LINGAM/schema-shield — Data engineering tool that classifies schema changes as SAFE, WARNING, or BREAKING so teams know if a change is safe to deploy.
Retail Bill-Scanning OCR Tool: Extracts P&L-style data from uploaded receipt images with WhatsApp integration.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering — Computer Science · KLN College of Engineering *2020 · GPA 8.1*
AWS Certified Cloud Practitioner (2022) · Microsoft Azure Data Fundamentals
Awards: Best Project and Best Team Player (KAAR Technologies); Best Outgoing Student (KLN College).